Classical and Self-Supervised Hankel Filtering for Magnitude-Only Rydberg Channel Estimation

Abdullah Haatim

***Abstract*—Rydberg atomic receivers can estimate wireless channels from signal magnitudes measured in the presence of a known reference. Channel structure can help this estimation, but too much filtering can remove useful signal components. This paper studies classical and learned ways to control Hankel filtering under the same nonlinear observation model. We examine hard rank constraints, array size and repeated projection, then develop a small estimator that learns smooth filtering from withheld measurements without true-channel training targets. Paired simulations show lower average error than the tested one- and four-sweep rank-search methods. A stronger classical shrinkage method has slightly lower average error and similar processing time. The learned estimator has fewer errors above the stated severe-error threshold, but this ranking reverses at a lower threshold. Weak references, few-path channels and signals far outside the training range expose further limits. The results show when structural filtering helps and why average accuracy, error thresholds and processing cost must be considered separately, rather than identifying a uniformly best estimator.**

***Index Terms*—Channel estimation, Hankel matrices, magnitude measurements, Rydberg receivers, self-supervised learning.**

I. Introduction

RYDBERG atomic receivers sense radio-frequency electric fields through highly excited atoms. Array models make it possible to study multiuser communication with these receivers [1], [2]. In the magnitude-only setting considered here, the received signal combines with a known reference before its amplitude is measured. Estimating the complex channel is then a nonlinear inverse problem. The reference provides phase information indirectly, but noise, pilot design and initialization still affect recovery.

A geometric channel also has structure across the array. A small number of propagation paths produces a sum of spatial exponentials, whose Hankel lift has low rank. This observation suggests inserting a structural step into a measurement-fitting algorithm. The practical questions are how strongly to filter, how often to apply the filter, and whether choosing its settings costs more than the resulting accuracy gain justifies.

This paper addresses these questions in one framework. We first study Hankel-structured expectation-maximization Gerchberg–Saxton estimation (HS-GS), which selects a hard rank using withheld pilots. We then replace per-channel rank search with a compact learned controller. The controller chooses smooth filtering and mixing strengths from the measurements and the current estimate. It is trained by predicting withheld observations, without true-channel targets. Both estimators retain the exact Rician observation model.

The paper makes three contributions. First, it measures when hard-rank filtering helps and when rank selection or repeated filtering removes useful channel detail. Second, it develops a 188-parameter estimator that learns the strength of a smooth Hankel filter from withheld observations, while keeping the exact Rician measurement update. Third, it tests whether learning is worthwhile against hard-rank and stronger classical shrinkage controls on paired observations. The result is an accuracy–failure–cost trade-off, not a universal winner: the learned estimator does not beat in-loop OptShrink on average dB error, but has fewer severe errors in the tested low-SNR cell.

A. Relation to Existing Methods

EM-GS already exploits the referenced magnitude model in atomic MIMO reception [1]. Xu et al. study channel estimation using a strong-reference linearization [3]; our experiments keep exact magnitude observations. URformer already unfolds phase retrieval for Rydberg receivers [4]. Thus neither the observation model nor unfolding alone is a new contribution. Spatial methods such as Quantum-MUSIC also show the relevance of array structure to quantum wireless sensing [5].

Alternating rank and structure projections are established [6], as are low-rank Hankel recovery [7], [8], structured alternating least squares [9] and data-driven singular-value shrinkage [10]. Learned Hankel reconstruction [11], self-supervision through withheld measurements [12], [13], and Rician-likelihood training [14]

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also have precedents. Self-supervised denoising has been studied for Rydberg sensor signals [15]. Our narrower contribution is to learn a small spatial correction within exact referenced-magnitude channel estimation and assess its value against these structural alternatives. We do not claim the first use of any of these ingredients or superiority over published URformer, for which a matched comparison is not available here.

Other relevant methods make different structural assumptions. LRPR-VBL uses a hierarchical low-rank matrix prior and variational EM for magnitude-only recovery [16]. HVaF exploits a Vandermonde factorization to recover exponential signals from incomplete samples [17]. LRHD studies low-rank Hankel denoising [8]. Our in-loop OptShrink control is not a reproduction of those full methods. Adapting them to the known-reference Rician model, with independent parameter selection, remains an empirical comparison gap. Their published denoising or recovery guarantees are not transferred to our nonlinear reconstruction.

II. Observation Model and Available Information

An array with N elements receives P pilots from K users. Its observations are

$$Z = |GS + B + W|, \qquad W_{np} \sim \mathcal{CN}(0, v), \quad (1)$$

where $G \in \mathbb{C}^{N\times K}$ is the unknown channel, $S \in \mathbb{C}^{K\times P}$ contains known pilots, and B is a known reference field. Magnitudes act entrywise. Noise samples are independent, with total complex variance v and variance $v/2$ in each real component. Receiver conversion gain is absorbed into the scaling. This effective measurement model does not simulate atomic dynamics or optical readout.

For a half-wavelength uniform linear array, each user channel is

$$g_{nk} = \sum_{\ell=1}^{L_k} \alpha_{\ell k} e^{-j\pi n \sin\theta_{\ell k}}, \qquad n = 0, \dots, N-1. \quad (2)$$

Unless specified otherwise, $\alpha_{\ell k} \sim \mathcal{CN}(0, 1/L_k)$ and angles are independent and uniform on $(-\pi/2, \pi/2)$. Pilots follow $\mathcal{CN}(0,1)$. The simulated reference is a broadside field with constant waveform and zero phase,

$$B_{np} = 10^{\mathrm{RSR}/20}, \qquad v = K10^{-\mathrm{SNR}/10}. \quad (3)$$

SNR is relative to aggregate user power, whereas RSR is relative to one user's power. At $K = 3$, RSR 12 dB corresponds to 7.23 dB relative to total signal power. Channels have unit expected power per user; individual realizations are not rescaled to identical energy. Reference calibration and noise variance are assumed known in the main comparisons.

A. Rician Likelihood and Local Identifiability

For $a = |\mu|$, a single observation has density

$$p(z \mid a, v) = \frac{2z}{v} e^{-(z^2+a^2)/v} I_0(2za/v), \quad z \ge 0. \quad (4)$$

With fixed v, a stable negative log-likelihood, up to observation-only terms, is

$$\ell(z, \mu, v) = \frac{(z - |\mu|)^2}{v} - \log I_{0e}(2z|\mu|/v), \quad (5)$$

where $I_{0e}(x) = e^{-x} I_0(x)$ for $x \ge 0$. Scaled Bessel functions avoid overflow at large arguments. If variance is estimated, the omitted $\log v$ term must be restored.

For one array row, let $\mu_p = g^T s_p + b_p$ and $x = [\mathrm{Re}\, g^T, \mathrm{Im}\, g^T]^T$. Where $\mu_p \neq 0$, the real Jacobian of its noiseless magnitudes has row

$$A_p = [\mathrm{Re}\, c_p^T, -\mathrm{Im}\, c_p^T], \qquad c_p = \mu_p^* s_p / |\mu_p|. \quad (6)$$

If A has rank $2K$, a nonsingular square sub-Jacobian and the inverse function theorem give local recovery. Therefore $P \ge 2K$ is necessary for this regular full-rank condition, but is not sufficient. A small minimum singular value means that recovery can still be unstable. This is a local statement for an unrestricted channel row, not a global or sparse-channel recovery guarantee [18].

With $B = 0$, a common phase rotation of g leaves its magnitudes unchanged. A known nonzero reference can break this ambiguity, but need not do so for every pilot/reference design. For example, if $b_p = c^T s_p$, then rotating $g + c$ leaves the observations unchanged. Unknown common reference phase also permits simultaneous rotation of channel and reference: $|e^{j\phi}GS + e^{j\phi}B| = |GS + B|$. Estimating an unrestricted common phase jointly with G cannot fix the absolute phase from these observations alone. An additional phase anchor or other information is required. The known-reference assumption is therefore substantive, not just a normalization.

B. Information References

For independent measurements, the real Fisher information is

$$J = A^T \,\mathrm{diag}(w_p) A,$$
$$w_p = \mathbb{E}\left[\left(-\frac{2a_p}{v} + \frac{2Z_p}{v}\frac{I_1(2Z_p a_p/v)}{I_0(2Z_p a_p/v)}\right)^2\right]. \quad (7)$$

The full channel information combines array rows in the corresponding real-coordinate ordering. One amplitude observation contributes an outer product and hence at most one information direction. Several different pilots and the known reference are needed to distinguish the complex channel coordinates.

Where the information matrices are nonsingular, the unconstrained reference is J^{-1}. The geometric constrained Cramér–Rao bound (CCRB) is [19]

$$C_{\text{geom}} = U(U^T JU)^{-1}U^T. \tag{8}$$

Here U is an orthonormal basis for the channel manifold's real tangent space at the true path configuration. It is constructed from derivatives with respect to real and imaginary path gains and spatial frequencies. With $\omega = \pi \sin\theta$, the complex derivative columns are $e^{-jn\omega}$, $je^{-jn\omega}$ and $-jn\alpha e^{-jn\omega}$. Stacking their real and imaginary parts defines the tangent space; gains and frequencies remain unknown parameters. The true model order is used only for this ideal reference.

We normalize the summed traces by the same summed channel energy used for NMSE. The gap between the two bounds describes the value of geometric information under local unbiasedness assumptions. It is not guaranteed attainable improvement. Our estimators can be biased, so these curves are not absolute lower limits on their measured NMSE.

III. Classical Hankel Estimation

A. A Common Rician Measurement Update

Let $M = GS+B$ and $\kappa = 2Z|M|/v$. The conditional complex field and EM update are

$$Q = Z \odot \frac{I_1(\kappa)}{I_0(\kappa)} \odot \frac{M}{|M|}, \tag{9}$$

$$G_{\text{EM}} = (Q-B)S^H(SS^H)^{-1}. \tag{10}$$

All ratios in the first line act entrywise. At $M = 0$, the continuous limit is zero; no substantial amplitude floor changes the likelihood. Linear solves evaluate the second line, requiring full row rank of the fitting pilot matrix.

For $\mathcal{L}(G) = \sum_{np} \ell(Z_{np}, [GS+B]_{np}, v)$, its Wirtinger derivative is

$$D = \frac{\partial \mathcal{L}}{\partial G^*} = \frac{(M-Q)S^H}{v},$$

$$G_{\text{EM}} = G - vD(SS^H)^{-1}. \tag{11}$$

Thus EM is an exact preconditioned gradient update for this likelihood. The corresponding packed real gradient is $2D$. This distinction fixes the factor of two when comparing gradient descent and EM-based unfolding.

All iterative comparisons start from spectral initialization. For row n, form C_n with rows $[s_p^T, b_{np}]$, let u_n be the leading eigenvector of $C_n^H \operatorname{diag}(z_n) C_n$, and set

$$c_n = \frac{\sum_p z_{np} |(C_n u_n)_p|}{\sum_p |(C_n u_n)_p|^2}. \tag{12}$$

The first K entries of $c_n u_n e^{-j \arg(u_{n,K+1})}$ initialize the channel row. Whenever pilots are withheld, initialization uses only the fitting subset. This rule is necessary to avoid leaking validation observations into an estimate.

B. Lifting, Projection and Rank Selection

For $g \in \mathbb{C}^N$, let $q = \lceil N/2 \rceil$, $m = N - q + 1$, and define

$$[\mathcal{H}(g)]_{ij} = g_{i+j}, \quad 0 \le i < m, \quad 0 \le j < q. \tag{13}$$

A channel with distinct spatial frequencies has Hankel rank at most $\min(L, m, q)$, with equality for nonzero gains and nondegenerate factors. The maximum rank is q. The inverse operation $\mathcal{H}^\dagger$ averages entries on each antidiagonal, so $\mathcal{H}^\dagger \mathcal{H}(g) = g$.

A hard-rank sweep lifts each user column, retains the largest r singular components, and averages antidiagonals. HS-GS applies Q_s such sweeps after each EM update. A finite number of these alternating steps does not guarantee a final relifted matrix of rank exactly r. We call them Cadzow sweeps rather than an exact projection onto the intersection of constraints.

The baseline uses the first 70% of pilot columns for fitting and the remaining 30% for selection. For each shared user rank $r \in \{1, \dots, q\}$, it fits the context pilots for T_{sel} updates and scores the held-out squared amplitude residual. The selected rank minimizes that score. The final estimate restarts from all pilots and runs T updates. Unless stated otherwise, $T/T_{\text{sel}} = 100/25$ and $Q_s = 4$. Rank q explicitly bypasses structural processing, recovering the paired EM-GS estimate with the same initialization and updates. The option to bypass does not guarantee that the selector chooses it whenever filtering is harmful.

To distinguish selection rules, we use *HS-amplitude* for this original baseline and *HS-Rician* when the candidate score is the held-out loss in (5). Both search settings separately for each test realization; all final fits use every pilot. The shared scalar rank is a computational choice, not a claim that users have equal path counts.

C. Smooth and Data-Driven Classical Controls

Hard truncation is only one use of structure. A smooth control replaces each singular value by

$$\widehat{\sigma}_i = \frac{\sigma_i^3}{\sigma_i^2 + \tau^2}. \tag{14}$$

Its classical version selects a multiplier from $\{0, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$ using held-out amplitude error, with τ equal to that multiplier times the current RMS singular value. The scale is recomputed each sweep. Zero bypasses filtering. Candidate and final update budgets match HS-amplitude. This is an empirical filter, not OptShrink.

Our stronger control instead uses OptShrink weights [10] inside the same Rician EM iterations. For

candidate rank $r < q$, let $s_{r+1}, \dots, s_q$ be the discarded singular values and, for $d \in \{m, q\}$, define

$$\phi_d(z) = \frac{1}{d-r}\left[\sum_{j=r+1}^{q} \frac{z}{z^2 - s_j^2} + \frac{d-q}{z}\right], \quad (15)$$

$$\mathcal{D}(z) = \phi_m(z)\phi_q(z),$$

$$\widehat{s}_i = -2\mathcal{D}(s_i)/\mathcal{D}'(s_i), \quad i \le r. \quad (16)$$

Other weights are zero. The implementation rescales the spectrum and sets weights at unresolved poles to zero. After antidiagonal averaging, a mixing factor blends the result with the EM update. Hankel lifting and nonlinear reconstruction create dependent errors; the original matrix-denoising optimality guarantees are not asserted for this adaptation.

OptShrink settings are chosen offline on independent development observations, separately by known noise level. The search includes ranks 1–15, mixing factors $0.25, 0.5, 1$, iteration budgets $10, 20, 50, 100$, and a no-filter option. Each candidate fits 21 pilots and is scored on nine withheld pilots using Rician likelihood. Settings are frozen before test generation. A final-only variant is tuned separately using the same grid. Inference uses all pilots and a noise-level lookup, without per-channel rank search. This distinction matters when comparing computation with HS-GS.

IV. Self-Supervised Structural Control

The learned estimator performs ten stages. Each stage fits the observations and then applies one smooth Hankel correction. The same small controller is shared across users and stages. Its interface takes only (Z, S, B, v); true channels and path counts are evaluation data, not estimator inputs.

A. Two Measurement Cores

The primary core relaxes the EM update:

$$G_{\mathrm{dc}} = G + \gamma_t(G_{\mathrm{EM}} - G), \quad \gamma_t = 2\,\mathrm{sigmoid}(a_t). \quad (17)$$

The secondary core uses $G_{\mathrm{dc}} = G - 2\eta_t D$, with

$$\eta_t = \gamma_t \frac{v}{2\|S\|_2^2}. \quad (18)$$

Both initialize $\gamma_t = 1$ and learn ten scalar update parameters. This comparison changes update geometry while retaining the exact likelihood. Validation likelihood selects the primary core before examining test channel errors.

B. Observable Controller and Structural Step

For each $H_k = \mathcal{H}(g_{\mathrm{dc},k})$, let $s_H = \|H_k\|_F/\sqrt{q}$ and $p_i = \sigma_i/\sum_j \sigma_j$. An 8–16–2 network with SiLU activation produces outputs (u, w). Its inputs are

$$\begin{aligned} f_1 &= \exp(-\textstyle\sum_i p_i \log p_i)/q, \quad f_2 = p_1, \\ f_3 &= \textstyle\sum_{i=q-\lceil q/2\rceil+1}^{q} p_i, \\ f_4 &= \log\!\left(1 + \frac{\mathrm{mean}[(Z - |G_{\mathrm{dc}}S + B|)^2]}{v}\right), \\ f_{5,k} &= \log\!\left(1 + \frac{v\,\mathrm{tr}[(SS^H)^{-1}]}{K\widehat{p}_k}\right), \end{aligned} \quad (19)$$

where $\widehat{p}_k = \|g_{\mathrm{dc},k}\|_2^2/N$, together with $P/(2K)$, $\log q$, and $(t+1)/10$. The residual mean is over all fitting observations; $t = 0, \dots, 9$. Positive denominator guards handle zero scales. These features describe the current spectrum, residual, noise, pilot support, size and stage. During training, P here is the context-pilot count.

The outputs set

$$\tau = s_H[0.001 + \mathrm{softplus}(u)], \quad \beta = \mathrm{sigmoid}(w), \quad (20)$$

$$F_\tau(H) = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \quad (21)$$

$$g_k^{t+1} = g_{\mathrm{dc},k} + \beta_{t,k}[\mathcal{H}^\dagger F_{\tau_{t,k}}(H_k) - g_{\mathrm{dc},k}]. \quad (22)$$

Equation (21) implements (14); the matrix is normalized by s_H before the solve for numerical stability. Reconstruction gradients pass through this solve, without an identity approximation to the backward pass. Spectral features still require singular values. Threshold and gate start at approximately $0.1 s_H$ and 0.1. The network has 178 parameters; adding ten update parameters gives 188.

The implementation floors channel power and the singular-value sum at 10^{-12}, and squared Hankel RMS scale at 10^{-24}. Pilot-Gram systems use direct solves, not a hidden ridge or pseudoinverse. The classical Bessel ratio uses scaled functions and its large-argument expansion above 10^4; the learned implementation uses a small-argument series when $\kappa^2 < 10^{-4}$. OptShrink sets a weight to zero at a relative squared spectral gap of at most 10^{-13}. These implementation-specific rules stay fixed during evaluation. The supplement records their implementation and checks.

The gate and threshold can compensate for each other. Neither is an estimated rank or a guarantee of safe filtering. To measure actual intervention, we also record

$$c_{t,k} = \frac{\|g_k^{t+1} - g_{\mathrm{dc},k}\|_2}{\|g_{\mathrm{dc},k}\|_2 + \epsilon}. \quad (23)$$

In the trained models, the gate is often near one: adaptation mainly changes the shrinkage strength rather than learning a reliable on/off rule.

C. Training and Deployment

Each training realization randomly assigns 21 of 30 pilot columns to a context set A and nine to a held-out set V. Only context observations enter initialization, features and the ten updates. Training minimizes

$$\mathcal{L}_{\mathrm{hold}} = \frac{1}{N|V|} \sum_{n,p \in V} \ell\big(Z_{np}, [\widehat{G}_A S_V + B_V]_{np}, v\big). \quad (24)$$

Masks are resampled each epoch. Independent validation cases use fixed masks; checkpoints are selected by equally weighted condition means of this held-out likelihood. True channels are not used for the loss or checkpoint selection. At deployment, weights stay fixed and all available pilots enter the estimator. There is no per-channel training or rank search. Held-out likelihood remains a prediction criterion, not a guarantee of minimum channel NMSE.

Training on 21 input pilots and deploying on 30 changes the controller's input distribution, including its pilot-count feature and measurement uncertainty. This choice uses every available pilot at inference; it does not establish invariance to the split. Pilot-count transfer tests measure one consequence, but do not isolate this mismatch from the value of additional observations. A matched retraining study would be needed for that causal claim.

D. What the Updates Guarantee

For fixed τ, the singular-value multiplier in (14) lies between zero and one, so $\|F_\tau(H)\|_F \le \|H\|_F$. Also,

$$\|F_\tau(H) - H\|_F \le \sqrt{q}\,\tau/2, \quad (25)$$

since $\sigma\tau^2/(\sigma^2 + \tau^2) \le \tau/2$. Let D_H be diagonal with antidiagonal multiplicities, so $\|g\|_{D_H} = \|\mathcal{H}(g)\|_F$. Averaging is an orthogonal projection in the lifted space. Therefore the gated step does not increase this weighted channel norm, and its weighted correction is bounded by $\beta\sqrt{q}\tau/2$. These statements bound an individual output; they do not prove contraction between different inputs or reduced error relative to the true channel.

The EM surrogate is $\|Q - B - GS\|_F^2/v$. Its normal equations imply that relaxation reduces it by

$$\gamma_t(2-\gamma_t)\|(G - G_{\mathrm{EM}})S\|_F^2/v \ge 0 \quad (26)$$

for $0 \le \gamma_t \le 2$. Subsequent structural filtering need not decrease either the surrogate or the likelihood. No convergence or low-noise consistency guarantee follows for the full learned estimator; in particular, its threshold need not vanish as measurement noise vanishes.

An NMSE-based stopping rule would require the unknown true channel and is therefore not available at inference. Rejecting a correction when likelihood worsens could protect that objective, but would not prove lower channel error or recovery of the correct phase. No such rule is added or described as a safe bypass here. The norm bound also does not make cubic shrinkage statistically optimal for the dependent errors produced by Hankel lifting. The solve is smooth for positive threshold; ordered spectral features need not be differentiable at every singular-value tie.

V. Experimental Design

The evidence comes from separate studies. Methods within a study share observations, but results from different studies are not pooled. The classical study tests structural assumptions; the original learned study tests training choices; the stronger-baseline study compares frozen learned models with tuned classical filters. Unless a fixed path count is specified, L_k is independently uniform on $\{3, 4, 5, 6, 7\}$. All estimators within a trial share the channel, pilots, reference and noise. Table I identifies the main protocols. The classical geometric generator rejects circular frequency separation below 10^{-10}, steering-matrix singular-value ratio below 10^{-8}, and pilot singular-value ratio below 10^{-8}. Reported results apply to these screened samples.

A. Learning Settings and Controls

Training uses 16,384 cases and validation uses 2,048. SNR takes seven equally spaced values from -10 to 20 dB; training RSR is 5, 12 or 20 dB. Half the cases independently sample three to seven paths per user. The other half use a common count drawn from $\{2, 4, 8, 12, 16\}$. Validation equally weights 126 SNR/RSR/path-setting combinations. Adam uses learning rate 10^{-3}, batch size 32 and 20 epochs. Training is single precision; evaluation and mathematical checks are double precision.

For each measurement core, we train no filtering (ten learned steps), globally learned filtering (one threshold and gate per stage, 30 parameters total), and the adaptive controller. Three seeds per choice give 18 models. The three seeds are 9101, 9102 and 9103. They change initialization, batch order and masks while sharing training cases. The main campaign took 3.67 hours with overlapping workers, excluding data preparation and preflight; this is not serial training time.

The full learned evaluation has 32 conditions with 400 paired cases each. Two noise-input conditions reuse observations, giving 12,000 distinct worlds rather than 12,800 independent worlds. The stronger OptShrink comparison uses a separate development set of 224 worlds and 1,152 fresh test worlds. Neither study's error totals nor timing samples are pooled with the other. The learned models remain frozen in the stronger comparison.

TABLE I
SETTINGS FOR THE PRINCIPAL COMPARISONS. $K = 3$ IN THE LISTED STUDIES; SEPARATE TRANSFER TESTS VARY K. HS BUDGETS ARE FINAL/CANDIDATE UPDATES.

Study	Array/pilots; RSR	SNR or complexity	Paired trials	Structural settings
Classical main/array	$N = 8, 16, 32$; $P = 10, 30$; 12 dB	$-5, 0, 5, 10, 15, 20$ dB	400/cell	HS: 100/25, four sweeps
Effective rank	$N = 16, 32, 64$; $P = 20$; 10 dB	Random SNR $[-10, 20]$ dB	300–400/cell	HS: 100/25, four sweeps
Classical filter comparison	$N = 32$; $P = 30$; 12 dB	$L = 3, 7, 14$; SNR 0,10 dB	400/cell	Hard/smooth: 100/25, four sweeps
Sweep sensitivity	$N = 32$; $P = 30$; 12 dB	$L = 3, 7, 14$; SNR 0,10 dB	100/cell, fresh	1,2,4,8 sweeps; own selection
Learned main study	$N = 32$; $P = 30$; 12 dB	Seven SNRs from -10 to 20 dB	400/cell	Ten stages; HS-amplitude control
Stronger shrinkage study	$N = 32$; $P = 30$; 12 dB	Seven SNRs and two stress cells	128/cell, fresh	Frozen models; tuned OptShrink
Post-review hard baseline	Same as stronger study	Same seven SNRs and stress cells	Same 1,152 worlds	One sweep; own Rician rank selection

After review, we add one-sweep HS-Rician to those same 1,152 saved worlds. It retains the 21/9 split, ranks 1–16 and 100/25 update budgets, but selects its own rank using one sweep in each candidate and final update. Only held-out measurements select rank. The protocol was recorded before this additional run, but the bank had already been inspected. This is a matched post-review follow-up, not a new blinded confirmation. The learned weights and OptShrink settings are unchanged.

B. Error, Uncertainty and Timing

For method m and trial i, define $e_{m,i} = \|\widehat{G}_{m,i} - G_i\|_F^2$ and $d_i = \|G_i\|_F^2$. We use

$$E_m = 10 \log_{10} \frac{\sum_i e_{m,i}}{\sum_i d_i}, \tag{27}$$

$$\Delta_{\mathrm{med}}(a, b) = \mathrm{median}_i \left[10 \log_{10}(e_{a,i}/e_{b,i}) \right]. \tag{28}$$

The curve gain $E_a - E_b$ is a ratio-of-sums gain and need not equal the paired median. Positive gain favors method b. Learned summaries average per-seed dB scores and then condition scores equally. A severe error means $e_{m,i}/d_i > 1$; it is not a communication outage probability.

Pointwise 95% intervals use 4,000 paired whole-world bootstrap resamples, retaining all users and method outputs together. Learned intervals condition on saved models; variation across seeds is reported separately and does not create additional independent test worlds. Pooled random-SNR medians depend on the declared SNR distribution. There is no blanket claim of superiority from a pooled average.

CPU timing uses one Apple M2 thread, batch size one and double precision, including initialization, feature extraction, structural steps and rank search where used. Learned code uses PyTorch; classical code uses NumPy/SciPy. Offline tuning and training are separate. Comparisons describe these implementations rather than device-independent speed ratios.

VI. RESULTS

A. When Hard Hankel Structure Helps

At $N = 32$, $P = 30$ and RSR 12 dB, the paired-median HS-amplitude gain over EM-GS ranges from 2.54 to 2.97 dB across six SNRs, with mean 2.70 dB. The vertical NMSE gaps are 2.38–2.90 dB (Fig. 1). Projection is active in all 2,400 trials. At SNR 5 dB and above, EM-GS is within 0.108 dB of the Rician CRLB. The geometric CCRB is 6.95–7.06 dB below that reference, while HS-amplitude remains 4.37–4.85 dB above it. The curves do not establish efficiency or attainable headroom for these biased estimators.

The benefit grows with array size. Averaging two pilot counts and six SNRs, the paired-median gains are 0.236, 1.044 and 3.264 dB at $N = 8, 16, 32$. Corresponding ratio-of-sums gains are -0.095, 0.866 and 2.938 dB. The EM-GS average changes by only 0.120 dB. A larger array provides more spatial samples of the same path structure, whereas the unrestricted baseline also acquires more unknown coefficients. At $N = 8$, the maximum Hankel rank is only four, and projection is bypassed in 42.94% of trials. A small positive median alongside a negative total-error gain shows why both aggregation rules must be identified.

B. Effective Rank and Structural Bias

Path count alone does not describe how many components are important. For singular values of the true noiseless user Hankel matrix, define [20]

$$r_{\mathrm{eff}} = \exp(-\textstyle\sum_i p_i \log p_i), \qquad \rho = r_{\mathrm{eff}}/q. \tag{29}$$

For example, a spectrum dominated by two values can have lower effective rank than a spectrum with four

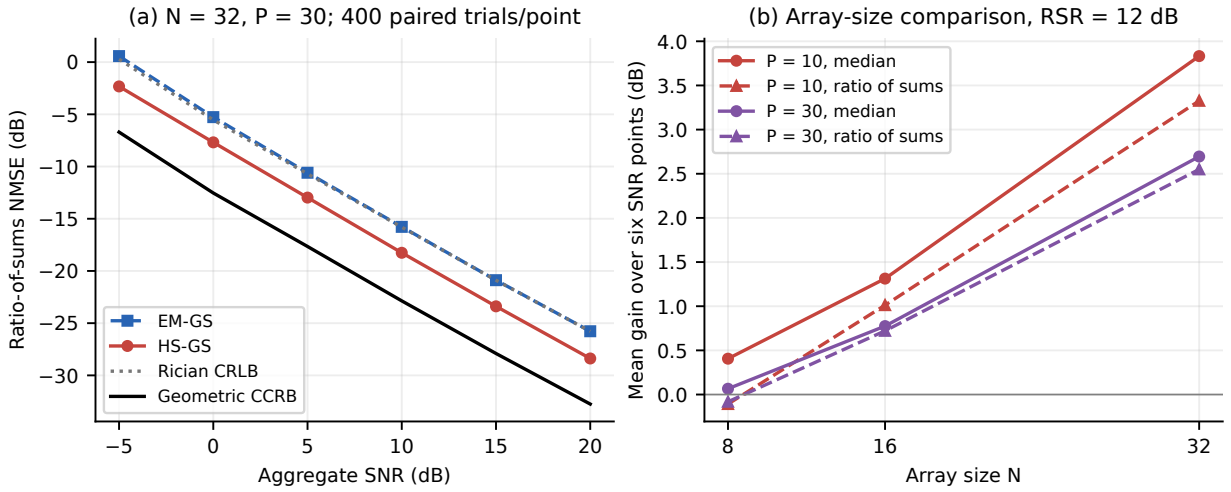


Fig. 1. Classical study. (a) NMSE and information references at $N = 32$, $P = 30$ and RSR 12 dB. (b) HS-amplitude gain versus array size for two pilot counts. The plot label HS-GS denotes HS-amplitude with four Cadzow sweeps. Panel (b) averages six SNR summaries; solid lines show paired medians and dashed lines show ratio-of-sums gains.

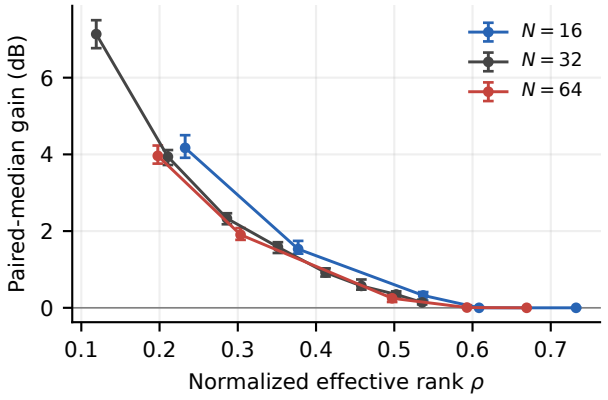


Fig. 2. Paired-median gain of the four-sweep HS-amplitude baseline against true-channel normalized effective rank. All three arrays use $P = 20$, RSR 10 dB and the same random-SNR distribution. Bars are pointwise paired-bootstrap intervals. This is a diagnostic trend, not an observable switching threshold.

similar values, even if both have algebraic rank four. We report the median ρ over users and worlds in a cell. This true-channel diagnostic is not supplied to either estimator.

Matched tests use $P = 20$, RSR 10 dB and SNR uniform on a millidecibel grid over $[-10, 20]$ dB. Path counts are $\{2, 4, 7, 9, 14\}$ for $N = 16$, even counts 2–16 for $N = 32$, and $\{8, 14, 29, 38, 48\}$ for $N = 64$. As ρ increases, the gain of the tested hard-rank estimator generally falls (Fig. 2). At $N = 32$, $\rho = 0.536$ gives median gain 0.141 dB with interval $[0.090, 0.211]$ dB. Thus the region near one-half of capacity marks a small benefit in this experiment, not a universal threshold.

At $N = 16$, $L = 14$ and $\rho = 0.732$, forcing rank seven gives median gain -0.154 dB, with interval $[-0.201, -0.111]$ dB. Adaptive selection has pooled median zero for these channels, but does not protect each trial. In a separate $N = 32$, $L = 14$, SNR 0 dB comparison, HS-amplitude is worse than EM-GS in 47.75% of 400 trials. Conversely, selecting below the algebraic rank need not be harmful: at $L = 7$ and SNR 0 dB, 78.75% select below seven, but only 8.75% of all trials are worse than EM-GS. Suppressing weak paths can help in noise.

C. Filter Strength and Placement Matter

The six-cell classical filter comparison uses $L = 3, 7, 14$ and SNR 0,10 dB, with 400 paired trials per cell. The smooth classical filter improves ratio-of-sums error over four-sweep HS-amplitude by 0.914, 1.382 and 1.413 dB at SNR 0 dB, and by 0.005, 0.623 and 0.773 dB at SNR 10 dB. The interval for the 0.005 dB result includes zero. This already shows that a gain over hard truncation need not require learning.

A fresh sweep study uses 100 worlds in each of the same six cells, with separate rank or threshold selection for one, two, four and eight sweeps. One-sweep hard processing improves over four sweeps in every tested cell, by 0.275–0.652 dB, with pointwise intervals above zero. More sweeps improve neither estimation nor computation automatically. These tests are separate from the learned comparison; the learned gain over the original four-sweep baseline is not evidence that it beats every hard-rank configuration.

Interleaving is also different from filtering once at the end. With the same four sweeps per application and a

TABLE II
LEARNED MAIN STUDY: SEVEN-SNR AVERAGES AND SEPARATE SNR 5 DB MEDIAN TIMING. LOWER NMSE IS BETTER.

Method	NMSE (dB)	Time (ms)	Severe (%)
Adaptive EM (188 weights)	-14.42	8.24	0.45
EM, learned steps only	-10.41	2.98	22.11
EM, global filtering	-12.43	5.61	19.19
Adaptive gradient core	-14.59	8.00	0.85
HS-amplitude, four sweeps	-12.74	525.62	14.25
EM-GS, 100 updates	-10.28	17.01	23.57
Rician gradient, hard Hankel	-11.06	47.39	20.93

TABLE III
FRESH STRONGER-BASELINE STUDY: SEVEN-SNR AVERAGE.

Method	NMSE (dB)	Severe (%)
HS-Rician, four sweeps	-12.890	13.06
HS-Rician, one sweep	-13.314	12.95
Final-only OptShrink	-14.374	3.79
In-loop OptShrink	-14.462	5.02
Adaptive EM, frozen	-14.319	0.89

shared selected rank, interleaving gains 0.204 dB, with interval $[0.169, 0.242]$ dB, averaged over six per-SNR medians on 300 paired trials per SNR. With one sweep per application, the corresponding gain is 0.239 dB, with interval $[0.204, 0.282]$ dB. These comparisons hold the rank fixed using interleaved four-sweep candidates; they do not match total computation or separately tune every placement.

D. Learning Versus Simple Learned Controls

The adaptive EM estimator improves over learned step sizes alone and globally learned filtering (Table II). Its mean gain over HS-amplitude is 1.675 dB, with interval $[1.618, 1.730]$ dB; seed-specific gains range from 1.533 to 1.787 dB. At SNR 20 dB the gain is only 0.139 dB, with interval $[-0.017, 0.294]$ dB. The gradient core has slightly better test NMSE, but EM remains the primary model because validation likelihood selected it before test evaluation. The hard-Hankel gradient control uses development-selected rank ten and one sweep per update.

The original timing uses 100 cases and three repeats at SNR 5 dB. Adaptive EM takes median 8.24 ms versus 525.62 ms for HS-amplitude. The roughly 64-fold ratio includes full per-channel rank selection and applies only to that comparator. It does not demonstrate that learning is faster than an offline-tuned classical filter.

E. Comparison With In-Loop OptShrink

The stronger classical control reverses the average-error ranking. In-loop OptShrink has 0.143 dB lower average NMSE than adaptive EM, with paired interval $[0.086, 0.198]$ dB (Table III). The learned method gains 0.723 dB at SNR -10 dB but loses 0.796 dB at 20 dB (Fig. 3). Selected OptShrink uses ten iterations through SNR 10 dB and twenty at 15 and 20 dB.

TABLE IV
POST-HOC ERROR-THRESHOLD SENSITIVITY ON THE STRONGER BANK. RATES AVERAGE SEVEN SNR CELLS; POSITIVE DIFFERENCES FAVOR LEARNING.

NMSE threshold	Learned (%)	In-loop OptShrink (%)	OptShrink minus learned 95% interval (points)
0.5	17.00	15.85	$[-1.935, -0.409]$
1	0.89	5.02	$[3.051, 5.283]$
2	0.00	0.00	—

The added one-sweep HS-Rician control reaches -13.314 dB, improving on four sweeps by 0.424 dB. Learning retains a 1.004 dB average gain, with paired interval $[0.922, 1.088]$ dB. This closes the one-versus-four-sweep gap on the same bank, not the comparison to every possible hard-rank method. In the separate weak-reference and two-path cells, one sweep instead beats learning by 1.976 and 1.144 dB.

The learned method has fewer severe errors: 0.89% versus 5.02% averaged over the seven SNRs. All occur at -10 dB in this sample, where rates are 6.25% and 35.16%. Mean per-case linear NMSE favors learning, 0.197 versus 0.215, but this is a different aggregation and does not reverse the primary average-dB result. Reporting both explains the average-error versus failure-frequency trade-off without changing the primary metric after observing results.

At the original threshold $e_i/d_i > 1$, the seven-SNR reduction in failure rate versus in-loop OptShrink is 4.13 percentage points, with paired interval $[3.051, 5.283]$. At -10 dB alone the interval is $[21.354, 36.979]$ percentage points. Learned seed-specific rates there are 5.47%, 5.47% and 7.81%, versus 35.16% for OptShrink.

The conclusion depends on the error threshold (Table IV). At threshold 0.5, OptShrink has fewer exceedances; at threshold 2, neither method has any observed exceedance. Zero observed events do not establish zero population risk. These exploratory, pointwise intervals retain paired worlds and fixed models and are not adjusted for multiple thresholds. Learning therefore does not dominate the whole error distribution. The original severe-error definition and primary NMSE metric are unchanged.

The post-review timing run measures all five methods together on the same independent 24-world SNR 5 dB timing bank, with three repetitions. Mean times are 5.16 ms for one adaptive model, averaged across three trained models, 5.68 ms for in-loop OptShrink, 3.87 ms for final-only OptShrink, and 119.08/358.07 ms for one-/four-sweep HS-Rician. Initialization and setting

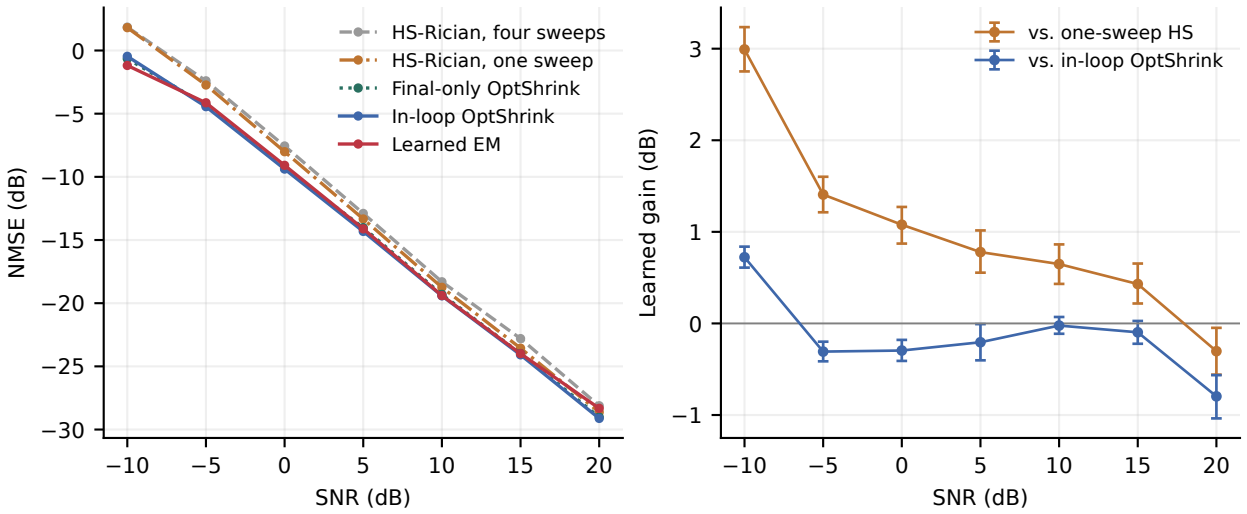


Fig. 3. Stronger classical comparison: 128 paired trials per SNR and frozen learned models. The one-sweep HS-Rician result is a post-review addition on the same saved worlds. Right: learned gains over one-sweep HS-Rician and in-loop OptShrink; positive favors learning. Intervals are pointwise and condition on the saved models.

selection or lookup are included; model loading and file I/O are excluded. These are new paired timing measurements, not averages with the earlier timing run. Backend differences remain. Learning avoids the hard method's per-case search, but has no large speed advantage over offline-tuned OptShrink.

F. Operating Limits and Additional Checks

Against HS-amplitude, learning loses 1.617 dB at RSR 0 dB, 1.104 dB for two paths and 0.203 dB for three paths (Fig. 4). These losses prevent a claim that the controller reliably identifies every channel on which filtering helps. The stronger study also finds no universal winner. At RSR 0 dB, selected OptShrink reaches -5.026 dB versus -7.069 dB for learning, but longer 50- and 100-update OptShrink runs reach -8.790 and -8.981 dB. Transferring a noise-dependent stopping rule can fail when reference strength changes. At two paths, HS-Rician reaches -17.833 dB, learning -16.814 dB and selected OptShrink -16.407 dB.

Input-removal tests train on 4,096 cases for 12 epochs with three seeds and evaluate two fresh banks totaling 2,304 worlds. Full inputs improve average error over spectrum-only, noise/residual-only and size/stage-only controllers. However, they lose 0.39–0.40 dB to noise/residual-only inputs at SNR 5 dB and 0.82–0.93 dB on two-path cases. Both information groups can help the average without improving every condition. A separately trained final-only learned filter is about 0.70 dB worse on its follow-up study's declared average; that experiment has its own training and test data and does not replace the classical shared-rank placement comparison.

Frozen-model dimension tests cover $N \in \{8, 16, 32, 64\}$, $K \in \{2, 3, 4\}$ and $P \in \{10, 20, 30\}$ one factor at a time. The gain interval at $N = 8$ includes zero. A smaller fresh pilot-count study, with 32 worlds per cell, extends to $P = 6 = 2K$. At SNR 5 dB, learned NMSE is -2.197 dB with six pilots and -14.332 dB with thirty. Full local Jacobian rank in sampled nonzero-reference cases does not ensure useful finite-noise accuracy.

Array mismatch also affects structural benefit. In a separate classical check at $N = 32$, $P = 30$, SNR 5 dB and RSR 12 dB, independent Gaussian element-position errors increase in standard deviation from zero to 0.02λ. The reference remains calibrated and error is measured against the perturbed channel. HS-amplitude gain drops from 4.955 to 3.917 dB for three paths, and from 0.264 to 0.050 dB for twelve paths; the latter interval includes zero. These are model-mismatch simulations, not hardware measurements.

A nearly noiseless check is especially instructive. At SNR 60 dB, outside the training range, and RSR 12 dB, the learned model reaches -6.647 dB versus -65.715 dB for EM-GS with 100 updates on 32 worlds. Disabling only filtering gives -44.020 dB; starting the full learned model at the true channel still gives -6.653 dB. These post-hoc diagnostics show that filtering itself can spoil an accurate estimate. They do not define a validated switching policy.

G. Retained Training and Calibration Evidence

A separate reduced-budget study trains 25 models for 300 epochs in total and tests 3,584 fresh worlds across

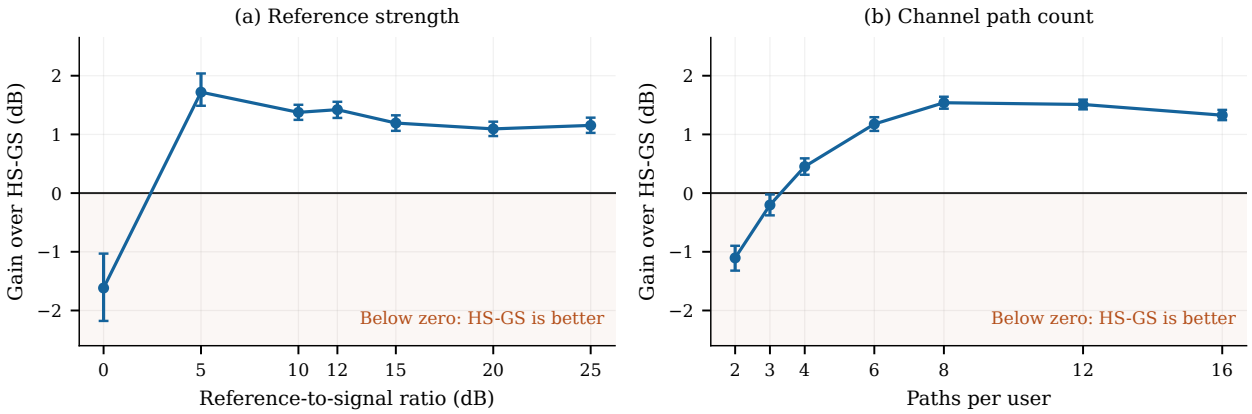


Fig. 4. Limits in the original learned study at SNR 5 dB. (a) Reference strength for three-to-seven-path channels. (b) Fixed path counts at RSR 12 dB. Gains compare adaptive EM with four-sweep HS-amplitude (HS-GS in the plot). Bars are conditional pointwise paired intervals; tested path counts overlap training.

56 SNR/reference/path conditions. Its training mixture includes RSR 0 dB, unlike the original study. Relative to its ten-stage controller, five stages lose 1.977 dB, while twenty gain only 0.181 dB and nearly double mean processing time. Increasing width from 16 to 32 gains 0.065 dB; two experts gain 0.023 dB; curriculum training gains 0.206 dB but loses 2.353 dB in its worst cell. No variant improves every cell. Three independent training datasets give two-initialization baseline means of -12.450, -12.124 and -12.229 dB. These are limited replications, not precise population uncertainty. The original primary models are not replaced using these test results.

Calibration sensitivity was also measured, not only discussed. In a 32-world SNR 5 dB check, assumed constant reference-phase errors of $\pm 10^\circ$ worsen learned NMSE by 2.36–2.47 dB; ramps from zero to $\pm 10^\circ$ worsen it by 0.78–0.84 dB. Supplying variance 6 dB too high causes a 3.05 dB loss. A separate 192-world diagnostic estimates a shared positive reference gain and noise variance while keeping phase supplied. It corrects some amplitude errors but worsens one tested cell by 1.048 dB, and adds 45.05 ms of preprocessing. It does not resolve unknown phase or drifting atomic response. The accompanying supplement retains these protocols, full tables, feature ablations and selected/final checkpoint checks; none is pooled with the primary comparison.

VII. Discussion and Conclusion

The two parts of the study lead to the same practical lesson: structural processing must be assessed against the error it removes, the bias it introduces and the work required to choose it. Low effective rank helps explain hard-projection gains, but effective rank is not an observed receiver input here. A rank-selection bypass, a learned gate and norm-nonincreasing filtering each fall short of a per-trial protection guarantee.

The learned method is most relevant when repeated estimation near a known operating population can reuse offline training. Its gain over global filtering supports measurement-conditioned control on the declared average. It does not establish that a neural network is necessary: offline-tuned in-loop OptShrink is competitive in time and slightly better in average dB error. Learning has fewer errors above the original threshold of one, but more above 0.5. This is a threshold-specific trade-off on the tested mixture, not generally better tail behavior or a safety guarantee.

For $m = N - q + 1$, one adaptive stage costs $O(NKP + NK^2 + K^3 + Kmq^2 + Kq^3)$ in the present implementation. Features still require an SVD, and reconstruction requires a dense solve. HS-GS repeats structural operations across candidate ranks and updates. A smaller parameter count alone is therefore not evidence of lower computational cost. Nor does applying the structure more often necessarily help: both the sweep study and the separately tuned final-only OptShrink control argue against that assumption.

The scope remains simulation-based estimation under an effective magnitude model with calibrated reference. The main confidence intervals exclude training-dataset uncertainty. Independent-data replications are limited; the new one-sweep comparison reuses an inspected bank. No matched URformer, LRPR-VBL, HVaF or full LRHD experiment is reported, so their accuracy and runtime relative to this controller are unresolved. There is no global recovery or full-estimator convergence theorem and no hardware validation. Non-Gaussian noise, outliers, two-dimensional arrays, wideband propagation and unknown phase drift are outside the tested model. Noise-variance mismatch alone is not a test of robustness

to a different noise distribution.

We have presented classical and self-supervised Hankel estimation within a common Rician framework. Hard structure helps when the channel leaves enough unused Hankel capacity, but rank selection and repeated projection can introduce avoidable error. A small controller can learn useful smooth filtering without channel targets and avoid per-channel search. Against a stronger classical shrinker, its contribution is a measured error–failure trade-off rather than uniform superiority. This distinction is necessary when deciding whether learned processing is worthwhile for magnitude-only Rydberg reception.

A. Reproducibility

The accompanying supplement records numerical details and the retained follow-up studies. The local reproduction packages contain code, generation seeds, observations, selected and final checkpoints, source hashes, predictions and audit scripts. The new one-sweep comparison and paired tail analysis have a separate record, leaving the original studies unchanged. No public repository URL or archival DOI is claimed in this draft; a permanent code/data destination remains an author action before submission. The Overleaf package contains manuscript dependencies, not the full experimental datasets.

REFERENCES

[1] M. Cui, Q. Zeng, and K. Huang, "Towards atomic MIMO receivers," *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 3, pp. 659–673, 2025.

[2] T. Gong *et al.*, "Rydberg atomic quantum receivers for classical wireless communication and sensing," *IEEE Wireless Communications*, vol. 32, no. 5, pp. 90–100, 2025.

[3] B. Xu, J. Zhang, Z. Chen, B. Cheng, Z. Liu, Y.-C. Wu, and B. Ai, "Channel estimation for Rydberg atomic receivers," *IEEE Wireless Communications Letters*, vol. 14, no. 9, pp. 2957–2961, 2025, author manuscript: arXiv:2503.08985.

[4] J. Xiao, J. Wang, M. Zeng, H. Xu, X. Li, and A. Nallanathan, "Channel estimation for Rydberg atomic quantum receivers: Unrolled phase retrieval from holographic snapshots," *IEEE Signal Processing Letters*, vol. 33, pp. 1696–1700, 2026.

[5] H. Kim, H. Park, and S. Kim, "Quantum-MUSIC: Multiple signal classification for quantum wireless sensing," *IEEE Wireless Communications Letters*, vol. 14, no. 6, pp. 1623–1627, 2025.

[6] J. A. Cadzow, "Signal enhancement—a composite property mapping algorithm," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 36, no. 1, pp. 49–62, 1988.

[7] Y. Chen and Y. Chi, "Robust spectral compressed sensing via structured matrix completion," *IEEE Transactions on Information Theory*, vol. 60, no. 10, pp. 6576–6601, 2014.

[8] M. Yin and R. S. Smith, "On low-rank Hankel matrix denoising," *IFAC-PapersOnLine*, vol. 54, no. 7, pp. 198–203, 2021.

[9] D. Zachariah, M. Sundin, M. Jansson, and S. Chatterjee, "Alternating least-squares for low-rank matrix reconstruction," *IEEE Signal Processing Letters*, vol. 19, no. 4, pp. 231–234, 2012.

[10] R. R. Nadakuditi, "OptShrink: An algorithm for improved low-rank signal matrix denoising by optimal, data-driven singular value shrinkage," *IEEE Transactions on Information Theory*, vol. 60, no. 5, pp. 3002–3018, 2014.

[11] Y. Huang, J. Zhao, Z. Wang, V. Orekhov, D. Guo, and X. Qu, "Exponential signal reconstruction with deep Hankel matrix factorization," arXiv:2007.06246v4, 2021, accepted author manuscript, IEEE Transactions on Neural Networks and Learning Systems.

[12] B. Yaman, S. A. H. Hosseini, S. Moeller, J. Ellermann, K. Ugurbil, and M. Akcakaya, "Self-supervised learning of physics-guided reconstruction neural networks without fully sampled reference data," *Magnetic Resonance in Medicine*, vol. 84, no. 6, pp. 3172–3191, 2020.

[13] C. Luo, H. Wang, T. Xie, Q. Jin, G. Chen, Z.-X. Cui, and D. Liang, "Matrix completion-informed deep unfolded equilibrium models for self-supervised k-space interpolation in MRI," arXiv:2309.13571, 2023.

[14] C. S. Parker, A. Schroder, S. C. Epstein, J. Cole, D. C. Alexander, and H. Zhang, "Rician likelihood loss for quantitative MRI using self-supervised deep learning," arXiv:2307.07072, 2023.

[15] Z. Liu, Q. Ren, W. Yang, Y. Tong, H. Wang, Y. Zhang, R. Zhi, J. Xie, M. Jing, H. Zhang, L. Xiao, S. Jia, K. Tang, and L. Zhang, "Self-supervised learning with noisy dataset for Rydberg microwave sensors denoising," arXiv:2601.01924, 2026.

[16] K. Liu, J. Wang, Z. Xing, L. Yang, and J. Fang, "Low-rank phase retrieval via variational bayesian learning," arXiv:1811.01574, 2018. [Online]. Available: https://arxiv.org/abs/1811.01574

[17] J. Ying, J.-F. Cai, D. Guo, G. Tang, Z. Chen, and X. Qu, "Vandermonde factorization of hankel matrix for complex exponential signal recovery—application in fast nmr spectroscopy," arXiv:1809.00750, 2018.

[18] B. Gao, Q. Sun, Y. Wang, and Z. Xu, "Phase retrieval from the magnitudes of affine linear measurements," *Advances in Applied Mathematics*, vol. 93, pp. 121–141, 2018.

[19] P. Stoica and B. C. Ng, "On the Cramér–Rao bound under parametric constraints," *IEEE Signal Processing Letters*, vol. 5, no. 7, pp. 177–179, 1998.

[20] O. Roy and M. Vetterli, "The effective rank: A measure of effective dimensionality," in *Proc. European Signal Processing Conference (EUSIPCO)*, 2007, pp. 606–610.